



Determination of shut-in time of shale gas well: Implication from differences in pore structure and imbibition characteristics of marine shale reservoir, SW China

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Abstract

Shut-in after fracturing is one of the important technical means to improve the recovery of shale gas well. At present, there is no strong scientific basis to support the formulation of shut-in time after fracturing. Our work has established a method to determine the reasonable shut-in time of shale gas well after fracturing by combining laboratory experiment data and field production data. For shale reservoirs with well-developed pores in the study area, the maximum water imbibition per unit volume and equilibrium time are 0.018542 g/cm³ and 18 days, which are much larger than those shale reservoirs with micro fractures (0.00939 g/cm³ and 9.6 days). Shale core will not generate new MF after water imbibition, but will form dissolution pores and expand existed MF. The time for the width of MF to reach its maximum value after water imbibition for shale cores with developed pores is 14 days, which is greater than that in shale cores with developed MF (7 days). The permeability of shale reservoirs with developed pores decreases by 8.1% after water imbibition, while the permeability of shale reservoirs with developed micro fractures (MF) loses by 92.5%, indicating that the gas phase flow channel of the former is better, while the latter has water blocking the gas flow channel and needs to be discharged as soon as possible. So, the former requires longer shut-in time than the latter. The purpose of shut-in is to maintain the opening of artificial fractures and the stability of formation energy. The time when the formation closure pressure is equal to the bottom hole flow pressure is selected as the shortest shut-in time, the time when the volume of core water imbibition is the highest is selected as the maximum time, and the best shut-in time is when hydration promotes the MF width to expand to the maximum value. Therefore, shale gas wells with good MF development, the shut-in time is 3~9.6 days, and the optimum shut-in time is 5 days; shale gas well with good pore development, the optimum shut-in time is 12.5 days.

Keywords Shale gas well · Shut-in time · Imbibition · Hydration · Optimization · Marine shale

1 Introduction

Due to the shale gas reservoir with extremely low porosity and permeability characteristics, the commercial exploitation of shale gas requires long horizontal well technology and multi-stage hydraulic fracturing technology (Dan 2015; Hazra et al. 2024; Tan et al. 2018). During shale fracturing operations, tens of thousands of cubic meters of fracturing fluids was injected into the shale reservoir and induce a large number of artificial fractures to accelerate the production (Luo et al. 2024; Ma et al. 2024; Warpinski et al. 2005). After fracturing, it can be roughly divided into three consecutive stages: shut-in (dissolving bridge plug, communicating fracture networks), flowback (rapidly discharging fracturing fluid near the wellbore, evaluating gas well

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productivity), and drainage and production (reasonable production allocation and stable production) (Li et al. 2020). Among them, shut-in have a great impact on shale gas well flowback, production and enhanced gas recovery (Makhanov et al. 2013; Yan et al. 2015; Yaich et al. 2015; Liang et al. 2017; Jiang et al. 2018). At present, most of shale gas wells in China are shut in for 7–30 d after fracturing, and the determination of shut-in time largely depends on the experience of operators, without theoretical and reasonable basis (Wang et al. 2021a, 2022a; Zeng et al. 2021). Optimizing the shut-in time is the key to fracturing and soaking implementation, and further improving shale gas recovery.

The purpose of shut-in after fracturing is to increase the flow energy of the formation fluid using a large amount of fracturing fluid retained in the reservoir based on the coupling effect of fluid flow and chemical reaction, to enhance the seepage effect of gas-water displacement, and ultimately achieve the purpose of increasing gas production (Zhao et al. 2019). Currently, the methods for determining the shut-in time of shale gas wells are mainly divided into statistical methods from field production data, theoretical model calculation methods, and core imbibition experimental methods. The field data statistical method requires a long time of production testing and a large amount of shale gas well production data with similar geological conditions. Due to the strong heterogeneity of shale reservoirs (Wang et al. 2022b; Lu et al. 2023), there are significant differences in the flowback of shale gas wells, which cannot provide a reliable reference for other wells (Cheng 2012; Bertoncello et al. 2014; Eltahan et al. 2021). As for the theoretical model calculation method, the simulation boundary conditions are limited and the accuracy of model related parameters is low, resulting in significant differences between the simulation results and production practice (Fakcharoenphol et al. 2016; Edwards et al., 2018; Shaibu et al., 2021; Zhou et al. 2022). The core imbibition experiment method determines the shut-in time by analyzing the length of imbibition equilibrium time (Taregh et al. 2022; Hu et al. 2022). Due to the rich clay content of shale reservoirs, a large number of micro fractures will occur after shale water imbibition, increasing the time for shale imbibition to reach the maximum amount (Makhanov et al. 2014; Wang et al. 2018; Zhou et al. 2022; Wijaya et al., 2021). Therefore, scholars focus on studying multiple factor imbibition models to obtain more reliable imbibition curves to determine the longest shut-in time (Lee et al., 2012; Wang et al. 2017, 2020, 2021b), ignoring the pore and micro-fracture changes of water rock action during imbibition process (Zhou et al. 2016; Zeng et al. 2022). And it is still controversial whether new pores and new micro fractures will occur in shale reservoirs during the imbibition process because fracturing requires hundreds of MPa to form a complex crack structure (Zhang et al. 2023).

It can be seen that the factors considered in determining the method of shut-in time at the current stage are relatively single and cannot fully meet the production requirements of shale gas wells. It is necessary to formulate a reasonable shut-in time from the perspective of combining on-site production and core experiments to promote the production increase of shale gas wells. In this work, the shortest shut-in time is determined based on the time when the bottom hole flow pressure equals the formation closing pressure to ensure that the proppant can be fully clamped after fracturing and to prevent artificial fractures from closing. The time when the water imbibition weight of the core reaches the maximum is regarded as the longest shut-in time to ensure sufficient gas-water replacement after fracturing of the shale reservoir. At the same time, by observing the evolution of pores and micro-fractures at different times after the shale imbibition water, the time when the micro-fractures width reaches the maximum value is calculated. This time is the optimal shut-in time, because shale gas has the strongest flow capacity. Comparing the length of these three times, the reasonable and optimal shut-in time of shale gas wells is determined.

2 Geological background, samples and methods

2.1 Geological background

China is rich in shale gas resources, and commercially exploited shale gas areas are mainly concentrated in the Sichuan Basin and surrounding areas. The Sichuan Basin is composed of three tectonic zones: the high steep tectonic zone of the southeast slope of Sichuan (including the high steep fold zone of eastern Sichuan and the low steep fold zone of southern Sichuan), the low steep tectonic area of the western Sichuan depression and the low-steep tectonic area of the central Sichuan uplift (Ma et al., 2019). The study area is located at the junction of the low steep fold zone in southern Sichuan and the low and gentle structural area of the central Sichuan uplift, and four structures are mainly developed in this area, including Mituochang anticline, Xishan anticline, Pulvchang anticline and Xiwenquan anticline, and the overall structural characteristics are northeast-southwest (Figs. 1a, b). Due to the widespread low-energy hypoxic environment caused by global sea intrusion, the Late Ordovician Wufeng Formation and the Early Silurian Longmaxi Formation formed a set of deep-water continental shelf facies black shale. The black organic-rich shale developed in the Wufeng Formation-Long 1 subsection is the main gas-producing section. According to the change of sea level lifting and rotation, the subsection of Long 1

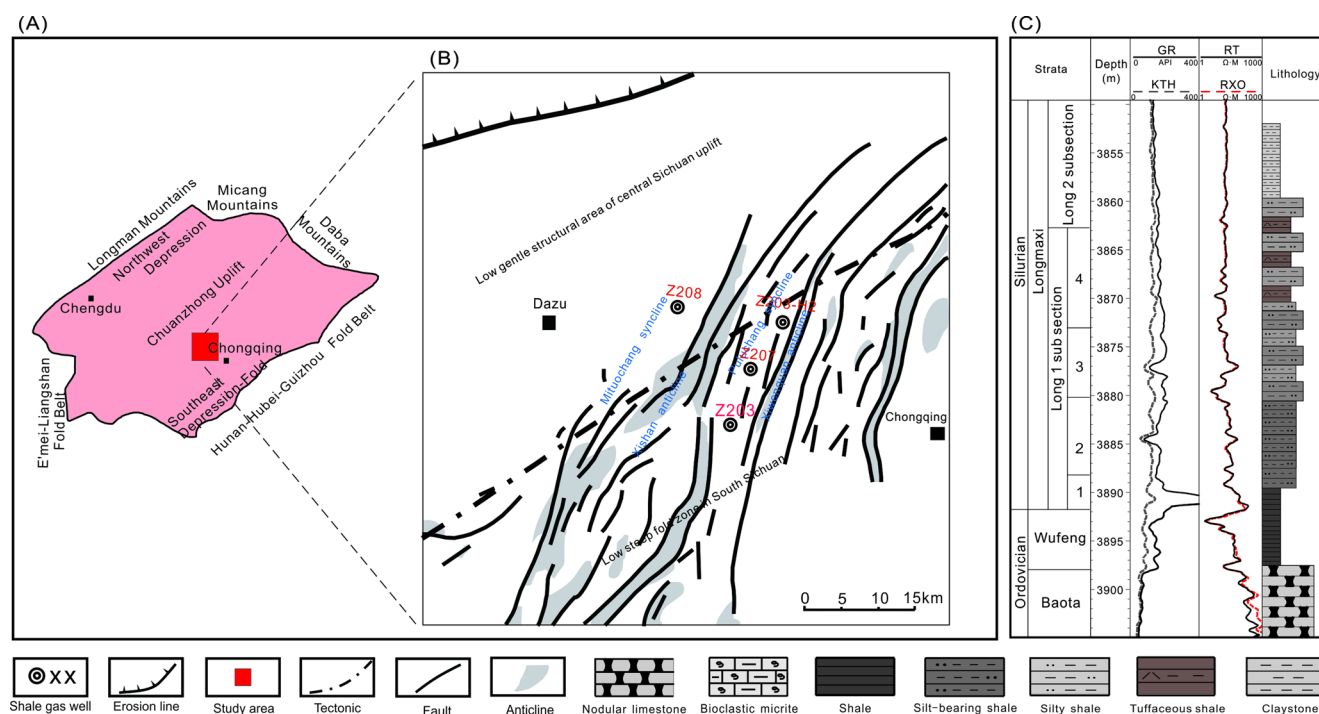


Fig. 1 Geographical map of the study area and comprehensive stratigraphic column

Table 1 General information of marine shale core samples of this study

Sample ID	Depth (m)	TOC (wt%)	Quartz (vol%)	Feldspar (vol%)	Carbonates (vol%)	Pyrite (vol%)	Clay (vol%)	porosity (%)
Z203-1	4103.92	4.78	60.73	7.40	12.07	3.18	12.62	5.78
Z203-2	4105.15	4.99	60.06	7.29	10.82	3.71	13.12	5.99
Z207-1	4384.71	4.53	60.22	6.47	11.38	4.20	14.73	4.48
Z207-2	4386.73	4.38	59.12	5.57	13.68	3.23	13.40	4.82
Z208-1	4366.00	4.56	61.19	6.02	11.45	4.12	15.22	6.03
Z208-2	4366.95	4.92	59.45	5.94	12.24	3.45	14.92	5.56

can be subdivided into four small layers: L_1^1 , L_1^2 , L_1^3 , L_1^4 , small layer (Fig. 1c). The targets for fracturing development are mainly distributed in the L_1^1 small layer. Therefore, the samples for the core experiment are mainly collected in this small layer.

The targets developed by shale gas well fracturing are mainly distributed in the L_1^{1-3} small layer. Therefore, our core experiments mainly focus on this section.

2.2 Samples

In this study, cores from Wells Z203, Z207, and Z208 were selected for argon ion polishing scanning electron microscopy (SEM), water imbibition, and permeability experiments. And basic parameters of core samples such as porosity, mineral composition, and TOC content have also been tested, as shown in Table 1. We can see that the core porosity of Well Z207 is about 1% lower than that of other wells, and other reservoir parameters are similar.

2.3 Methods

2.3.1 Imbibition weight and permeability monitoring

As shale gas is mainly developed through hydraulic fracturing, deionized water was applied to shale core (plug-sized) imbibition experiments to obtain the weight change curve with time after core water imbibition. An analytical balance (BSA224S-CW) with an accuracy of 0.0001 g was used to measure the core weight after different imbibition time. And all weight data were automatically measured by computer, reducing the error in operation. Meanwhile, the permeability of the plug-sized core is also tested during the imbibition process by steady-state method. Before testing permeability, it is necessary to use a sponge to wipe off the floating water on the surface of shale core. Considering that the permeability of shale will induce micro fracture changes after water imbibition, this permeability tester uses a hierarchical flowmeter test to enhance the testing range of permeability.

Refer to Fu et al. (2019) for specific imbibition experimental procedures.

2.3.2 SEM fixed-point observation during imbibition water

The pore and micro-fracture characteristics of shale samples after water (deionized water) imbibition different time were observed using FEI Quanta 650 FEG field emission scanning electron microscopy (SEM) which produced by FEI Czech Republic S.r.o. The resolution of the equipment under high vacuum conditions of 30 kV is 1.0 nm. Before water imbibition, the core samples were made into 1 cm × 1 cm cube, then the cube core surface was flattened by mechanical polishing and argon ion polishing. Finally, the sample surface was divided into equal 100 (10 × 10) small squares by laser etching, ensuring that the SEM can observe the same position after water imbibition. Before each SEM observation experiment, the sample should be dried at a low temperature of 60°C to prevent water vapor from polluting the sample bin during the electron microscope vacuum pumping process. After drying, gold plating is also required on the sample surface to ensure the resolution of SEM observation. The samples from Well Z203, Well Z207, and Well Z208 were conducted to fixed point observation experiment after imbibition water for different time (before water imbibition and 1, 3, 7, 14 days after water imbibition). The imbibition method is immersion imbibition and the immersion environment is normal temperature and pressure.

2.3.3 Field data analysis

Real-time monitoring of the wellhead pressure during shut-in process from shale gas wells in the studied areas of Longmaxi reservoir were conducted. After converting the wellhead pressure to the bottom hole flow pressure, the cross plot of the bottom hole flow pressure and the shut-in time was draw. The shortest shut-in time was determined by analyzing the relationship between formation closure pressure and bottom hole flow pressure. And the maximum shut-in time was optimized through the relationship between wellhead pressure derivative and shut-in time, combined with core imbibition experiments.

3 Results and discussion

3.1 Effect of micropore structure on the water imbibition of shale reservoirs

3.1.1 Micro SEM pore structure characteristics under dry conditions

The micro pore structure of shale reservoirs affects the imbibition and water-rock reaction of shale (Sui et al. 2018). The SEM image of shale core under dry conditions can effectively reflect the micro pore structure characteristics. Previous studies have shown that shale reservoir pores are mainly organic pores (OP) and develop in migrating organic matter (OM) (Zhao et al. 2016). So, this study focuses on the observation of OP in migrating OM and micro fractures (MF) of shale cores.

The OP of samples Z203-1 and Z208-1 exhibit a circular pore shape, with the maximum OP diameter of sample Z208-1 reaching 600 nm, which is larger than that of sample Z203-1 (Figs. 2a, c). However, the OP in the sample of Z207-1 is not developed (Fig. 2b). On the contrary, sample Z207-1 has more MF developed, with a maximum MF width of 3 μm and a MF length of 40 μm (Fig. 2e). The scale of MF in samples Z203-1 and Z208-1 is relatively small, and the MF width is only 700 μm, the MF length is also limited (Figs. 2d, f). Because sample Z207-1 MF are more developed, shale gas can migrate along MF and inorganic pores, reducing the pressure in OP (Fu et al. 2021a), resulting in underdeveloped OP and small porosity (Table 1).

3.1.2 Characteristics of water imbibition curve and influencing factors

The cross plots between water imbibition volume of shale core and imbibition time are shown in Fig. 3. The volume of water imbibition gradually increases with imbibition time, but the slope of the curves obviously decreases slowly (Figs. 3a–c). For samples from Well Z203 and Well Z208, the increase in imbibition volume is very small, accounting for only 4.33% of the total water imbibition volume after the imbibition time exceeds 400 h (Figs. 3a–c). But for Well Z207, the increase in water imbibition volume is less than 5% only after 200 h (Fig. 3b). In addition, the maximum imbibition volume of Well Z207 is 0.00939 g/cm³, which is much smaller than Well Z203 and Well Z208, with the maximum imbibition volume of 0.0161442 g/cm³ and 0.018542 g/cm³, respectively. There are various reasons for the difference in imbibition volume, mainly including porosity, pore structure, water saturation, clay mineral content, etc. (Dehghanpour et al. 2013; Yang et al. 2016; Ge et al. 2015). Due to the use of dry samples for water imbibition experiments in this

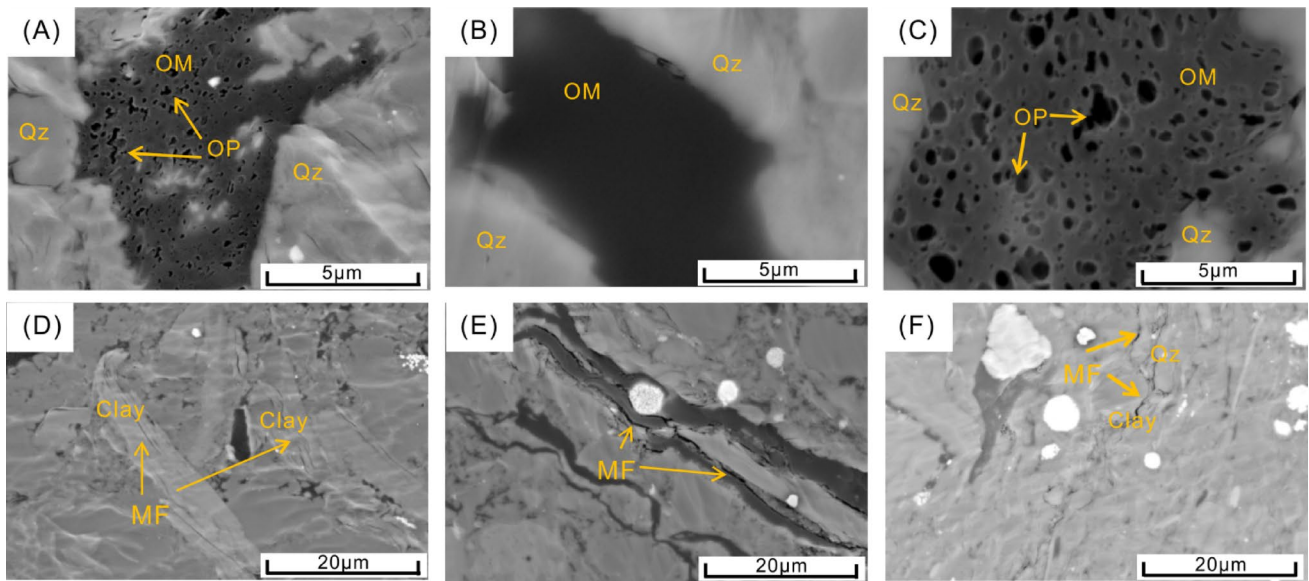


Fig. 2 Characteristics of OP and MF in shale cores. The OP characteristics of shale samples from Z203-1, Z207-1, and Z208 are respectively shown in (a–c). The MF characteristics of shale samples from Z203-1, Z207-1, and Z208 are respectively shown in (d–f)

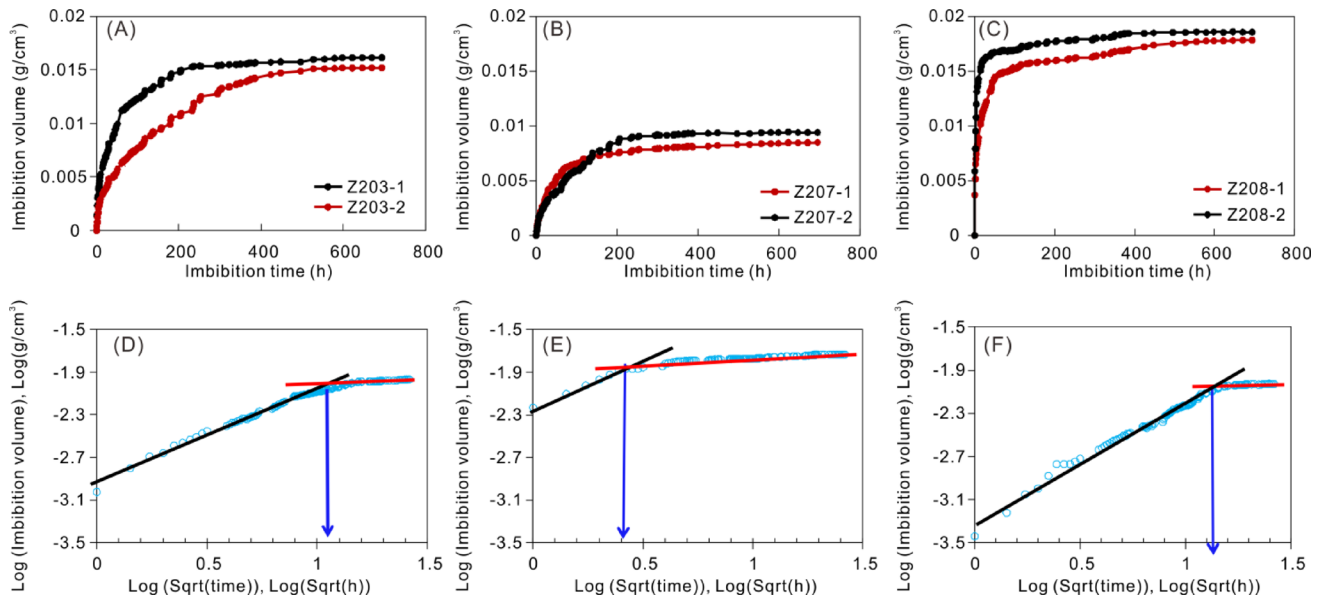


Fig. 3 Water imbibition profiles of shale cores of well Z203, Z207 and Z208 were shown in (a–c). And the logarithmic relationships between time and imbibition weight were shown in (d–f)

study, the original water saturation can be ignored. These three wells come from the same area, and there is not much difference in mineral composition (Table 1). Therefore, the main reasons for the low imbibition volume and short imbibition equilibrium time of Well Z207 are porosity and pore structure. Its porosity is only 4.65%, which is much smaller than that of Well Z203 (5.89%) and Well Z208 (5.80%). Moreover, shale reservoir of Well Z207 is characterized by poor OP development, small OP size, and poor connectivity (Fig. 2), making it difficult to imbibe water. This results in low water imbibition volume. However, the high

degree of development of MF in the shale reservoir of Well Z207 results in a shorter time for it to reach equilibrium in terms of water imbibition (Figs. 3d–f). The following uses mathematical equations to further analyze the differences in imbibition characteristics.

If the influence of gravity is not considered, imbibition water volume of per unit length of rock core can be expressed using the following equation (Handy 1960):

$$Q_w^2 = \left(\frac{2A_c \varphi k_w S_w P_c}{\mu_w} \right) t = A \cdot t \quad (1)$$

Where A_c is the surface area, φ is the porosity, P_c is the capillary pressure, k_w is the permeability of water, S_w is the front water saturation, μ_w is the viscosity of water and t is the imbibition time, A is a constant.

Both sides of the equation are simultaneously squared and logarithmic, and Eq. (1) can be rewritten as:

$$\log f(Q_w) = \log \left(\sqrt{\left(\frac{2A_c \varphi k_w S_w P_c}{\mu_w} \right)} \cdot \sqrt{t} \right) = \frac{1}{2} \sqrt{A} \log(t) \quad (2)$$

It can be found that a positive proportional relationship between the imbibition volume and the square root of time from Eq. (1). A general observation indicates that each imbibition curve tends to be divided into three parts: a linear imbibition part, a transition part, and a plateau part (Yang et al. 2017). The plot of the relationship between the cumulative imbibition volume and imbibition time on the log-log coordinate shows a Fischer diffusion type behavior (Fick 1995), where the percentage of water imbibition steadily increases with $t^{1/2}$ at the initial stage and then tends to stabilize near fluid saturation. Therefore, we can analyze the imbibition time corresponding to the maximum value of shale imbibition water on a logarithmic coordinate. The point where the two slopes intersect is the equilibrium time of imbibition water (Figs. 3d, f), and the equilibrium time of water imbibition in the three wells was determined. Imbibition equilibrium time of Well Z207, Z203 and Z208 is 230 h, 405 h, and 430 h respectively. Due to the poor development of micro pores in Well Z207, the maximum volume of water imbibition is lowest. The time for imbibition to reach equilibrium is shortest because of the better development of MF.

3.2 Effect of water imbibition on pore and micro-fracture expansion in shale reservoirs

3.2.1 Characteristics of pore changes at different water imbibition time

Loucks et al. (2012) classified the pore types of shale reservoirs into OP, inter granular pores and intra granular pores of matrix minerals. This study conducted SEM fixed-point observation experiments on shale samples of different days after water imbibition, summarized the influence of hydration on different types of pores, and then guided the optimization of shut-in time for shale gas wells.

For sample Z208-1, we can observe that there is no significant change in the OM and OP and after water imbibition (Figs. 4a–d). This is because the geochemical characteristics of OM itself determine that it doesn't have sufficient contact with water. In addition, OP are mostly nanoscale pores. Related studies have shown that most OP are oil wetted (Su et al. 2018; Liu et al. 2018), making it difficult for water to

enter these pores during the imbibition process. However, due to expansion of inorganic minerals after water imbibition, they compress the OM, causing the OP at the edges of the OM to shrink or deform (Fig. 4d).

Changes in inorganic pore morphology and volume are mainly caused by hydration after shale imbibition water (Qian et al. 2017; Gupta et al. 2017). Matrix pores, as channels for water imbibition and diffusion, play an important role in hydration and also serve as the starting point for hydration and dissolution. The SEM fixed-point observation of inorganic pore morphology in shale after water imbibition reveals that mineral dissolution occurs from the edge of mineral particles or pore edges. With the increase of imbibition time, the dissolution of hydrophilic mineral particles intensifies and ultimately leads to the detachment of mineral particles, forming larger inorganic pores (Fig. 4). For calcite with strong solubility, complete dissolution process or particle falling off can be seen (Figs. 4e–h). After hydration, complex pore networks can be formed, and some particles may detach due to enhanced hydration (Fig. 4h).

We performed image processing on the acquired SEM photos to obtain shale surface porosity. It was found that the increase in shale surface porosity before three days was not significant. On the 7th day of spontaneous water imbibition, the surface porosity of the shale increased rapidly, approximately 15 times that of the dry state. By the 14th day of water imbibition, the surface porosity still increased, but the increase significantly slowed down (Fig. 5a). It is worth noting that Well Z207 with poor pore structure has a smaller increase in surface porosity after imbibition compared to Well Z203 and Well Z208 with good pore structure. And after 7 days of water imbibition in Well Z207, the surface porosity remains basically unchanged because of Well Z207 MF development. During the water imbibition process, MF can ensure rapidly contact between water and shale reservoir rocks, accelerate hydration, and complete changes in pores and MF in a short period of time. However, the pore evolution during the water-rock reaction in the Well Z203 and Well Z208 shale reservoirs with improved pore structures is relatively slow.

3.2.2 Characteristics of MF changes at different water imbibition time

After different days of water imbibition, through fixed-point observation with SEM, it was found that hydration did not produce new MF. However, it can be observed that the original MF in shale underwent significant changes in both width and length during the hydration process (Figs. 4m, p). Due to the enhancement of hydration, the width of MF increases with the increase of imbibition time, reaching its maximum after 7 or 14 days of water imbibition (Figs. 4o,

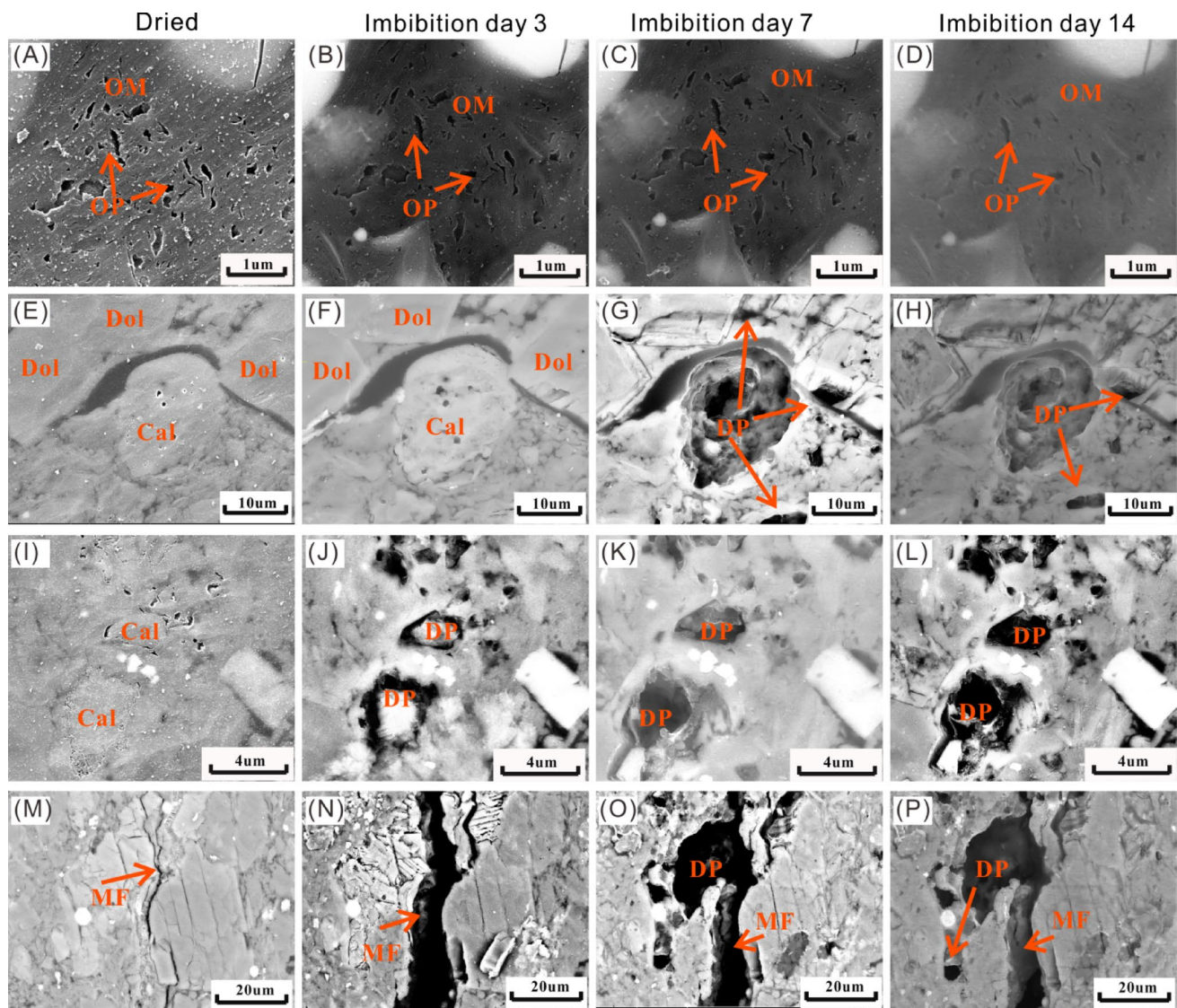


Fig. 4 Microscopic pore structure characteristics of shale sample Z208-1 before and after water imbibition. The characteristics of OP changes before and after water imbibition are shown in **a-d**. The characteristics of calcite changes before and after water imbibition are

shown in **e-h**. The characteristics of intergranular pore changes before and after water imbibition are shown in **i-l**. The characteristics of MF changes before and after water imbibition are shown in **m-p**

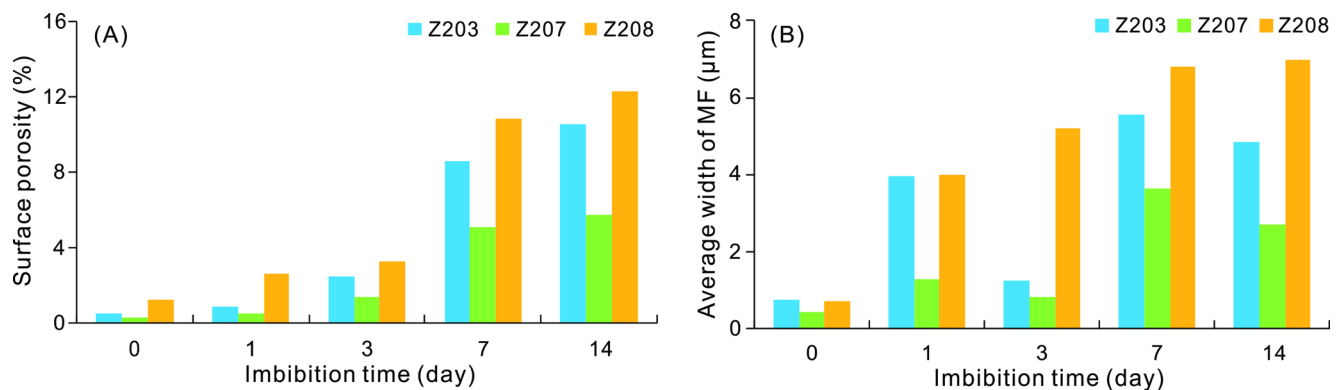


Fig. 5 **a** Surface porosity and **b** average width of MF changes after imbibition water different time

p). At the same time, the mineral dissolution at the edge of MF causes particle detachment, which can increase the width and length of MF. Compared with the original state, the width of MF after hydration can increase by several to ten times. Overall, it can be seen that the shale does not generate new MF after water imbibition, but rather expands on the basis of the existing MF, while generating a large number of dissolution pores, thereby increasing the permeability of the shale.

Through statistical analysis of the average MF width of water imbibition at different times, it was found that the overall trend of average MF width increasing with water imbibition (Fig. 5b). The average MF width of Wells Z203 and Z207 reached its maximum value on the 7th day after water imbibition, and there was a slight decrease on the 14th day. However, the width of MF in the Z208 Well continues to increase. After the water imbibition exceeds the 7th day, the increase in MF width significantly slows down. The reason for the decrease in the width of MF may be: (1) mineral dissolution or particle detachment to seal the MF; (2) continuous expansion of clay minerals (Sui et al. 2018).

After the shale reservoir of Well Z207 imbibes water, the increase in surface porosity and MF width is smaller than that of Well Z203 and Well Z208. The Well Z208 with the best pore structure has the greatest increase. From this, it can be seen that shale reservoirs with better pore structures are more conducive to the formation of dissolution pores and the expansion of MF through water rock reaction and hydration.

3.3 Effect of water imbibition time on permeability of shale reservoirs

In this study, the permeability at different water imbibition time was tested. Figure 6 shows the evolution of permeability after spontaneous water imbibition experiments. The first data point of per permeability curves is the absolute permeability in dry condition. The second data point represents the permeability immediately after imbibition water. At the initial stage of imbibition, the permeability of all samples decreases due to water filling in the pores and MF. Overall, the permeability of all shale samples after water imbibition is lower than that of the dry state. The permeability of the two shale samples from Well Z207 decreased by 92.5% and 99.6% after water imbibition, respectively. Samples Z203-1 and Z203-2 lost 30.1% and 86.9% of their base permeability respectively. While the samples from Well Z208 lost only 12.8% and 8.1% of its permeability. Most likely, this is because of the pore structure of shale reservoirs in Wells Z203 and Z208 is superior to Well Z207, leading to the absolute permeability of Wells Z203 and Z208 larger than Well Z207. During the imbibition process, water can be imbibed spontaneously into pores in reservoirs of Well Z203 and Z208. So, the MF are not filled with water which can facilitate gas flow. However, the pores of Well Z207 are not well-developed, and MF will be filled after water imbibition. The tested permeability is close to the matrix permeability (Chakraborty et al. 2017) and much lower than the other two wells. At the same time, after water imbibition of the shale gas reservoir with better pore structure, the contacted degree of the water and rock is more sufficient, and

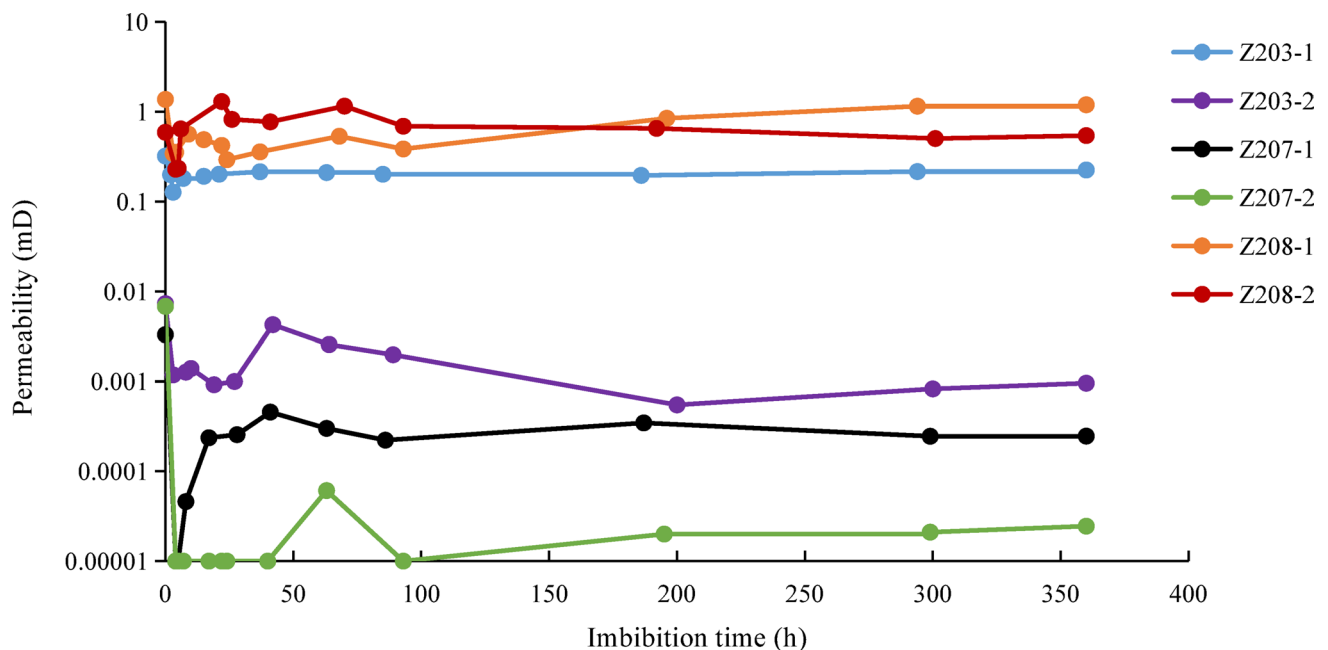


Fig. 6 Changes in shale permeability of Well Z203, Z207 and Z208 at different imbibition times

the stronger the water rock reaction is, the more conducive to the formation of dissolution pores and the expansion of MF (Sui et al. 2018; Yan et al. 2015). Therefore, we can see that the permeability loss of Wells Z203 and Z208 is much lower than that of Well Z207 after water imbibition (Fig. 6).

These results indicate that the absolute permeability at dry condition is the determining factor in predicting permeability response after different imbibition times, with higher permeability samples showing improvements, while samples with lower permeability showing little benefit, and even decreasing permeability over time. This understanding is consistent with the research findings of Chakraborty et al. (2017). It is important to mention that the permeability of all samples tends to stabilize after 200 h of water imbibition. But Wells Z203 and Z208 are improvements, while Well Z207 is decrease. The shut-in time of shale reservoir similar to Well Z207 should be shortened in favor of shale gas flow.

In the previous analysis, it was found that shale gas wells with different pore structures have different imbibition and pore variation characteristics after water imbibition, which have an important impact on shut-in time of shale gas well. However, from the perspective of laboratory experimental analysis, it cannot meet the needs of on-site production, and needs to be combined with specific shale gas well practical experience, especially the pressure changes during shut-in process and “sand production” after shale gas well opening.

4 Optimization of shut-in time

4.1 Implication of core experiment results on shut-in time

The purpose of analyzing the shut-in time of shale gas well after fracturing from the perspective of laboratory experiment is to keep the formation energy stable and the flow channel unblocked (Zhao et al. 2019; Zhou et al. 2022; Wijaya et al., 2021). This involves the evolution behavior of shale imbibition water volume and MF width after water imbibition. Because after shale reservoir fracturing, the larger the imbibition water volume, the greater the degree of contact between the fracturing fluids and the rock, which is more conducive to gas water replacement (Fu et al. 2021b). Meantime, after water imbibition, hydration helps to improve the seepage capacity of shale reservoir and is conducive to the recovery of shale gas (Makhanov et al. 2014; Wang et al. 2018).

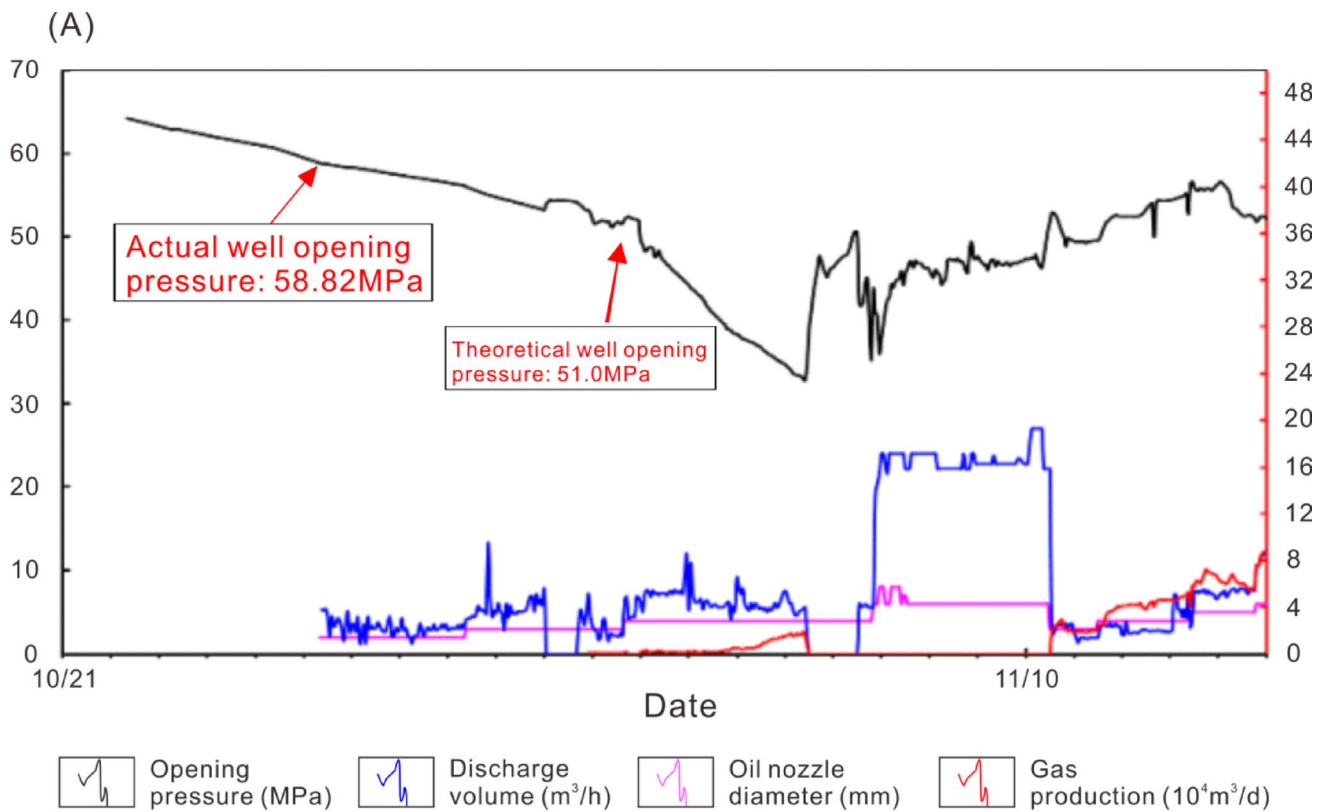
There are many factors affecting shale imbibition, including mineral composition, pore structure, porosity, fluid properties, etc. However, shale gas wells in the study area are located in the same area, where the sedimentary environment is similar, and the mineral composition of

shale reservoirs in the same formation is similar (Table 1). In addition, the fracturing plans and fracturing fluid systems between different wells are also similar. Therefore, the water imbibition volume and water imbibition rate of different shale gas wells in the study area are mainly controlled by the microscopic pore structure. From Sect. 3.1, we can see that the time for water imbibition to reach equilibrium of Well Z207 is shorter than that of Well Z203 and Well Z208 (Fig. 3) due to the poor development of pores, and water can enter matrix pores through MF and quickly balance under capillary pressure. However, the shale reservoirs of Well Z203 and Well Z208 with well-developed pores take a longer time to reach equilibrium of water imbibition. Therefore, shale reservoirs with more developed OP (e.g. Well Z203 and Well Z208) need longer shut-in time to achieve the maximum volume of imbibition fracturing fluids and replace more shale gas.

After fracturing, the shale rock will undergo complex physical and chemical reactions with the fracturing fluids to change shale reservoir pore structure (Zhou et al. 2016; Zeng et al. 2022). The surface porosity and MF width of shale gas Well Z203 and Well Z208 with good pore structure continue to increase after 7 days of imbibition water, while surface porosity basically unchanged and MF width decrease of Well Z207 with more developed MF (Fig. 5). This may be the rapid rate of water imbibition and water rock reaction due to the Well Z207 with MF development. After 7 days of imbibition, the dissolution effect is significantly weakened. Without increasing additional pore space, the continuous expansion of clay minerals may compress MF, resulting in a decrease in MF width. Therefore, shale gas wells with poor pore structure need shorter shut-in time to ensure smooth flow of shale gas.

4.2 Implication of field data on shut-in time

The determination of shut-in time by field shale gas well production data analysis is mainly based on the effective support of proppant and stability of wellhead pressure. Hydraulic fracturing technology is the main means for commercial development of shale gas wells at present (Dan 2015; Tan et al. 2018). In order to maintain the opening of artificial fractures and smooth flow of shale gas into the wellbore, proppant shall be added during fracturing process. If the shut-in time of shale gas well after fracturing is relatively short (the bottom hole flowing pressure is greater than the formation closing pressure), the proppant will flow back out of the wellhead with the fracturing fluids, causing the artificial fracture to close and decreasing gas production efficiency. Shale gas wells produce a large amount of proppant and have low production (Fig. 7). Therefore, it is necessary to regularly monitor the change law of the



(B)



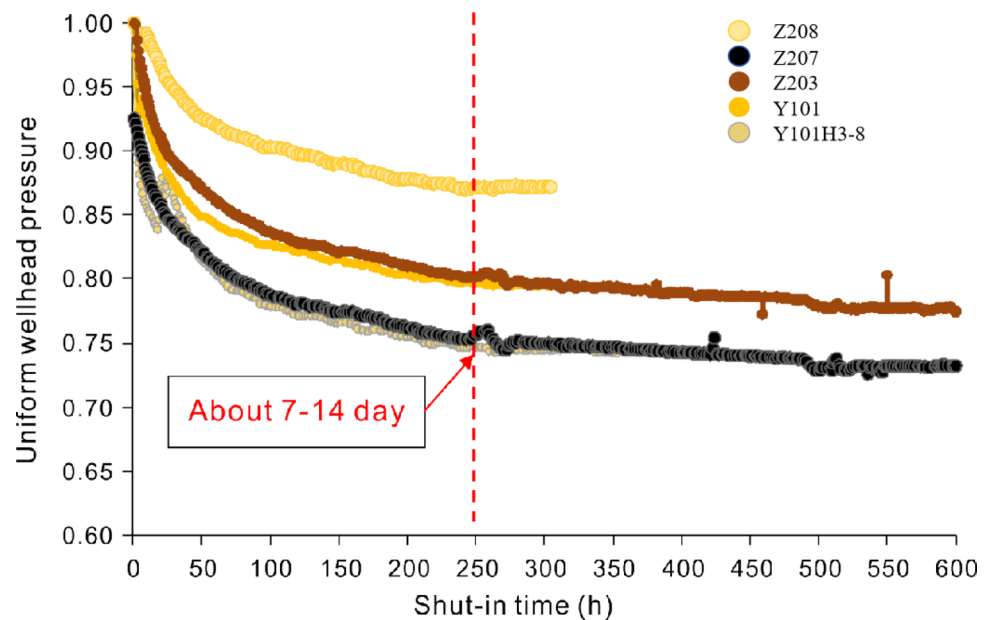
Fig. 7 **a** Typical well discharge and production curves; **b** Proppant flowback photos in the study area

wellhead pressure of shale gas wells to determine the time point corresponding to the formation closing pressure and bottom hole flowing pressure, and the shut-in time should not be longer than this time point.

Meantime, the rate of wellhead pressure drop characterizes the degree to which fracturing fluids imbibes into the fractures and matrix during the shut-in process. The greater rate of wellhead pressure drop, the higher the degree of imbibition fracturing fluids. Figure 8 shows that the inflection point of wellhead pressure changes occurs in 7–14 days

in the process of shut-in of shale gas wells in the study area and adjacent areas. This indicates that the shale reservoir fully imbibes fracturing fluids after 7–14 days, achieving MF expansion during the shale water rock reaction process. The shut-in time of shale gas well is extended to this time point, which helps to fully replace shale gas with fracturing fluids and improve the production of single well.

Fig. 8 Schematic diagram of well-head pressure change of shale gas wells during shut-in in process of the study area and adjacent area



4.3 Comprehensive analysis of shut-in time

The purpose of shut-in is to replace shale gas through reservoir imbibition and expand MF to promote shale gas flow and recovery (Liang et al. 2017; Jiang et al. 2018). Therefore, the displacement efficiency of shale gas by reservoir imbibition is mainly achieved by the equilibrium time when reaching the maximum volume of imbibition fracturing fluids. The greater the imbibition volume, the more shale gas will be replaced, which is more conducive to shale gas recovery. If the shut-in time exceed this time, there may be reservoir water lock or reservoir damage (Xue et al. 2022; Jiang et al. 2022). The shut-in promotes the flow of shale gas, including maintaining the opening of artificial fractures and the expansion of hydration micro fractures. To ensure the opening of the artificial fractures, it is necessary to fully clamp the proppant in the fractures. When the pressure in the fracture is less than the formation closing pressure, it can effectively ensure that the proppant will not be discharged back to the ground, offset part of the formation pressure, and keep the artificial fractures open to promote shale gas production. Therefore, the shut-in time should not be too short. Hydration promoting the expansion of MF in shale reservoirs is one of the key research directions in the process of shut-in (Zeng et al. 2021). The imbibition process is accompanied by complex hydration. On the one hand, it promotes the expansion of MF, and on the other hand, it continuously increases the imbibition volume to ensure the release of shale gas. Under the action of hydration, the width of MF will first increase and then decrease. The time when the width reaches its maximum value during the shut-in process is the optimal shut-in time. If the shut-in time is too long, the water saturation in the reservoir will be high,

which is not conducive to the flow of gas and water phases (Yao et al. 2023). Therefore, based on the experimental data (imbibition and water rock reaction) and field data (changes of bottom hole pressure with time during well shut-in), a scheme for optimizing the shut-in time of shale gas wells was established (Fig. 9).

In Fig. 9, the shut-in time should not be shorter than the time point corresponding to the bottom flow pressure equivalent to formation closure pressure (to prevent proppant flowback and artificial fracture closure), and should not be longer than the time point at which the displacement efficiency of water imbibition is maximum (to prevent reservoir water lock or reservoir damage). If the time corresponding to the maximum MF width under hydration is within the previous two time points, then this time point is the optimal time for shut-in. If the time corresponding to the maximum MF width under hydration is not within the previous two time points, then only the shortest and longest shut-in time need to be determined, which is the reasonable shut-in time.

In the study area, the burial depth of shale reservoirs is greater than 4000 m, and the formation closure pressure is relatively high, about 98 MPa (Sun et al. 2023). The value of static rock pressure plus wellhead pressure is less than the formation closure pressure within 3–5 days after shut-in. Therefore, the shut-in time in the research area should not be less than 5 days. For well Z208 with good pore development, the imbibition equilibrium time is about 430 h (about 18 days); for well Z207 with MF development, the imbibition equilibrium time is about 230 h (about 9.6 days) (Fig. 3). Therefore, the reasonable shut-in time of shale gas wells in the study area can be divided into two categories: one is shale gas wells with good MF development, and the shut-in time is 3–9.6 days; the other is shale gas well with

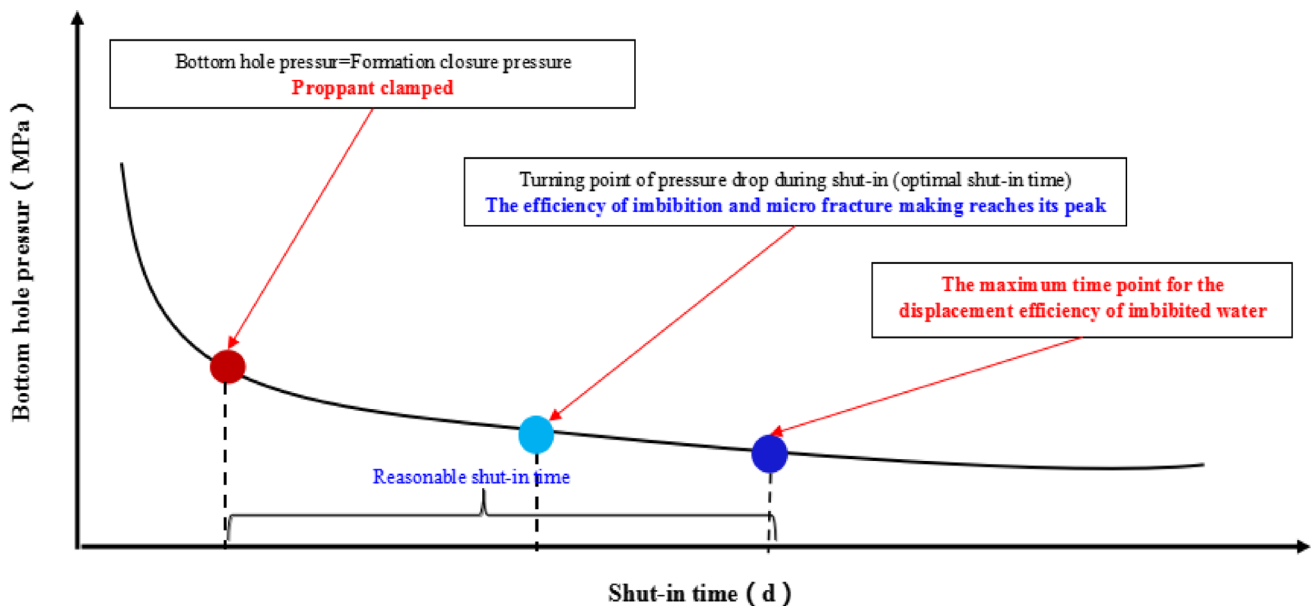


Fig. 9 Design chart of shut-in time technical scheme for shale gas well

good pore development, and the shut-in time is 5–18 days. In order to find the best time to shut-in for shale gas well, it is necessary to combine the permeability and MF width changes at different times of imbibition water. It takes about 3 days for the shale reservoir permeability of Well Z207 to reach its maximum value after decreasing due to water imbibition, while the maximum value of Well Z208 permeability after decreasing is about 12.5 days (Fig. 6). The imbibition time in Well Z207 corresponding to maximum width of MF after water imbibition is 9.6 days, while in Well Z208 it is 18 days (Fig. 4). The shorter the imbibition time, the smaller the volume of water imbibed into the core, resulting in a relatively high permeability. The phenomenon of mismatch between the imbibition time of the maximum permeability value and the maximum MF width value is caused. The optimum shut-in time for shale gas wells with good pore structure in the study area is about 12.5 days, while the shale gas wells with poor pore structure is about 5 days.

5 Conclusions

In our work, the weight, permeability, and changes in micro pore fracture patterns of shale cores are monitored as the water imbibition time increases. Meanwhile, the pressure variation law during the shut-in process of shale gas wells was recorded. Based on the comprehensive experimental test results and the pressure variation characteristics of shale gas wells, the shut-in time of shale gas wells has been optimized, and the following understanding has been obtained.

- (1) The volume and rate of shale reservoir imbibition water, the surface porosity, and the width of MF after water imbibition are controlled by the microscopic pore structure. Shale reservoirs with superior pore structure have larger imbibition volumes and smaller imbibition rates. After water-rock reaction and hydration, the increase in surface porosity and MF width is greater than that in shale reservoirs with poor pore structure.
- (2) Shale reservoir with well-developed pores in the study area have a high water imbibition volume (0.018542 g/cm^3), longer time (18 days) to reach the maximum water imbibition volume. After water imbibition, permeability lost only 8.1%, and surface porosity and MF width increase 20 times and 15 times, respectively. The time for MF width reaching maximum value is 14 days. The above parameters of shale reservoirs with developed MF and underdeveloped pores are different from shale reservoir with well-developed pores.
- (3) Based on the experimental data (imbibition and hydration) and field data (changes of bottom hole pressure with time during well shut-in), a scheme for optimizing the shut-in time of shale gas wells is established. Shale gas wells with good MF development, the shut-in time is 3–9.6 days, and the optimum shut-in time is 5 days; shale gas well with good pore development, the shut-in time is 5–18 days, and the optimum shut-in time is 12.5 days. This solves the misconception of using the same shut-in time in the same research area.

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Data availability The data that support the findings of this study are available on request from the corresponding author.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this manuscript.

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